Ideation Phase

Brainstorm & Idea Prioritization Template

DATE	TEAM ID
12 June 2026	
PROJECT NAME	**MAXIMUM MARKS**
E-commerce Application (ShopEZ)	4 Marks

> **Brainstorm & Idea Prioritization Template:**
> Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.
>
> Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.
>
> Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

STEP-1: TEAM GATHERING, COLLABORATION AND SELECT THE PROBLEM STATEMENT

1.1 Team Gathering & Collaboration Setup

The ideation session was initiated by assembling the engineering project team to address the technical and user-experience gaps present in conventional online retail applications. In alignment with academic guidelines for the Second Year Engineering Mini Project, collaborative channels were established using version control systems (GitHub), communication tools, and distributed whiteboarding spaces. Roles were distributed symmetrically to ensure thorough coverage of frontend interfaces, backend services, database orchestration, and data security measures.

1.2 Problem Statement Selection

Modern digital commerce platforms frequently suffer from fragmented performance under high concurrent loads, rigid filtering systems, and complex checkout sequences that elevate cart abandonment rates. Concurrently, small-to-medium administrators struggle with opaque inventory mechanisms and a lack of granular analytics.

Selected Problem Statement: How might we build a secure, high-performance, and intuitively navigable e-commerce web application ("ShopEZ") that balances a seamless, frictionless customer acquisition funnel with a robust, data-driven administrative dashboard for operational tracking?

STEP-2: BRAINSTORM, IDEA LISTING AND GROUPING

2.1 Brainstorming & Idea Listing

The team conducted a timed, volume-driven ideation session yielding numerous granular functional components. The core focus remained on architectural feasibility and user experience patterns suitable for university evaluation. The raw ideas generated include:

- Elastic multi-attribute search indexing with real-time autocompletion.
- One-click multi-tiered checkout pipelines to minimize transaction steps.
- Dynamic relational inventory tracking with automated low-stock alerts.
- State-managed client-side persistent shopping carts using localized storage.
- Admin analytical panels with embedded visualization tools for transaction reporting.
- Secure encrypted cryptographic user authentication and structural role-based routing.
- Paginated, relational product detail configurations supporting nested reviews.

2.2 Feature Grouping and Categorization

To structure the system architecture logically, the raw brainstorming outputs were grouped into unified clusters representing core application subsystems:

Subsystem Cluster	Brainstormed Core Features & Functional Elements
Customer Catalog & Discovery	Dynamic multi-tiered search, parameter-driven product filtering (price, category, ratings), paginated product detail displays, and threaded customer review structures.
Transaction & Cart Operations	Localized persistent shopping cart management, real-time quantity adjustments, state-driven cart persistence, and a multi-step checkout pipeline with automated tax/shipping calculations.
Administrative Operations	CRUD interfaces for complete inventory tracking, centralized order management tables, status mutation controls, and database-driven analytical reporting.
Security & Core Platform	Cryptographic hashing for user authentication tokens, role-based visual access controls (Admin vs. Customer), and relational database schema constraints.

STEP-3: IDEA PRIORITIZATION

The prioritized features were mapped systematically across an Importance vs. Feasibility matrix to delineate the Minimum Viable Product (MVP) boundaries for the academic term submission, filtering high-value core logic from complex secondary enhancements.

1. High Importance / High Feasibility (Core MVP - Immediate Execution):

- Relational database schema execution containing explicit integrity rules for products, users, and orders.
- Session-persistent shopping cart allowing dynamic addition, removal, and live total recalculation.
- Secure session management and login authentication parsing Admin and Customer interfaces.
- Unified backend inventory management forms (Create, Read, Update, Delete) for platform administrators.

2. High Importance / Low Feasibility (High Value - Simplified for Mini Project):

- Secure payment gateway integration (Simulated via a sandbox payment state-machine rather than live API processing).
- Real-time sales analytics dashboards (Executed using lightweight static embedded charts linked directly to SQL aggregate views).

3. Low Importance / High Feasibility (Secondary Focus - Post-MVP Polish):

- Multi-attribute product filtering and advanced elastic search parameters.
- Nested client-side interactive rating systems and customer feedback threads.